

Lê Tiến Phát

Computer Engineering
Embedded systems & IoT

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EDUCATION

Computer Engineering

Ho Chi Minh City University of
Technology

HCMUT, VNU-HCM
2022 - 2026

Studies completed.
Diploma pending.

TECHNICAL SKILLS

Programming

C/C++ · Python · SQL
MIPS assembly

Embedded & IoT

ESP32-S3 · Raspberry Pi
Tasmota · RS485/Modbus
MQTT

Tools & testing

Git · Linux · Pytest
Node-RED · Grafana
Fault injection
Troubleshooting

Analysis

Signal processing
Model evaluation · Qiskit
Technical documentation

Communication

English communication
Technical reporting

PROFILE

Computer Engineering studies completed at HCMUT; diploma pending.
Internship experience in IoT integration and firmware troubleshooting, with academic projects in edge AI, MIPS assembly and quantum machine learning experiments.

Seeking entry-level embedded, testing, manufacturing technology and technical implementation roles.

PROFESSIONAL EXPERIENCE

IoT Development Intern

Lotus Viet Nam Electronic Service - Trading -
Manufacturing Co., Ltd. | Jun - Aug 2025

- Integrated ESP32-S3 controllers and environmental sensors using MQTT, Node-RED and Grafana.
- Developed custom Tasmota drivers for RS485/Modbus devices; troubleshoot communication, configuration and integration faults.
- Tested continuous operation and sensor-data stability; documented configurations and troubleshooting procedures.

ACADEMIC & INDEPENDENT PROJECTS

SCADA / ICS Anomaly Detection & Edge Deployment

Academic capstone | HCMUT | 2026

- Built a signal simulator with replay and fault injection; investigated preprocessing errors and false alerts.
- Deployed TensorFlow Lite inference on Raspberry Pi and evaluated Isolation Forest on ESP32-S3 with emlearn, comparing detection performance, latency and hardware constraints.

Signal Filtering in MIPS Assembly

Computer Architecture coursework | 2025

- Developed a single-coefficient Wiener filter using integer arithmetic with decimal scaling, signal parsing and file I/O.
- Implemented coefficient calculation, filtered-output generation and mean-squared-error reporting.

github.com/lxalis/Computer-Architecture

Quantum Machine Learning Experiments

Four-person university project | 2025

- Ran Qiskit-based experiments and generated results for the final presentation; resolved legacy-library compatibility issues.
- Iterated with a teammate on code revisions and repeat testing; co-presented findings alongside the team leader.

DeskFlow - Workspace Booking & Pricing

Independent software project | 2026

- Built a FastAPI/SQLAlchemy backend and TypeScript client; added automated tests and investigated booking, timezone and revenue-rounding defects.

github.com/lxalis/Deskflow